Implementation of Binary to Gray Code Converter in Quantum Dot Cellular Automata

Generated complete project report.

Implementation of Binary to Gray Code Converter in Quantum Dot Cellular Automata

Project Identification

• Institution: CMR Institute of Technology, Hyderabad

• Department: Electronics and Communication Engineering

• Academic Year: 2024-25

• Guide: Dr. G. Rajender

• Members: Aleti Harneeth Reddy; Alladi Nithin Kumar; GVN Bharadwaj; Kavali Harshavardhan

Abstract

This project implements a Binary-to-Gray Code Converter using Quantum Dot Cellular Automata (QCA) design concepts.

Objectives

1. Study QCA as a nanoscale digital design technology.

2. Implement Binary-to-Gray code conversion logic.

3. Provide Verilog reference modules and testbenches.

Boolean Equations

2-bit

G1 = B1

G0 = B1 XOR B0

4-bit

G3 = B3

G2 = B3 XOR B2

G1 = B2 XOR B1

G0 = B1 XOR B0

4-bit Truth Table

QCA Design Concepts

• QCA cells represent binary information through electron configuration.

• QCA wires transfer cell polarization.

• Majority gates and inverters are fundamental QCA logic primitives.

• Clocking controls computation and signal propagation.

Workflow

Workflow ```text Binary Input | v Binary-to-Gray Logic | +--> MSB passed directly +--> Adjacent bits XORed | v Gray Code Output | v Simulation / Verification +--> XOR testbench +--> 2-bit exhaustive testbench +--> 4-bit exhaustive testbench ```

Project Structure

Repository Structure ```text Implementation-of-Binary-to-Gray-code-Converter-by-using-Quantum-Dot-Cellular-Automata/ ■■■ README.md ■■■ docs/ ■■■ verilog/ ■■■ reports/ ■■■ presentation/ ■■■ data/ ```

Source Materials

Supplied Source Materials - Main 55-page QCA Binary-to-Gray project report. - 18-slide mini-project presentation. - 7-page Verilog/acknowledgement document. - Separate 2024 GPS/GSM women-safety journal paper supplied alongside the project. The women-safety paper is preserved separately because it is a different project topic.

Verilog Files

- verilog/xor_gate.v
- verilog/xor_gate_tb.v
- verilog/binary_to_gray_2bit.v
- verilog/binary_to_gray_2bit_tb.v
- verilog/binary_to_gray_4bit.v
- verilog/binary_to_gray_4bit_tb.v

Verification

Gray = Binary XOR (Binary >> 1). The included testbenches exercise the reference logic.

Tools Mentioned

The source report references QCA Designer and Microwind Lite.

Deliverables

Generated PDF, DOCX, PPTX, XLSX and ZIP files are built from the committed project documentation and implementation files.

Truth Table

Binary	Gray

0000	0000
0001	0001
0010	0011
0011	0010
0100	0110
0101	0111
0110	0101
0111	0100
1000	1100
1001	1101
1010	1111
1011	1110
1100	1010
1101	1011
1110	1001
1111	1000